Hints: Probability – 4. The probabilistic method across mathematics

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $\operatorname{dist}(x, W)^2 = \|Px\|^2 = x^T P x$, where P is the orthogonal projection onto $W^\perp$.

Hint 2. $\mathrm{E}\, x^T P x = \operatorname{tr}(P\, \mathrm{E}[xx^T]) = \operatorname{tr} P$, since $\mathrm{E}[xx^T] = I$. Answer: $n - k$.

Problem 2

Note. If $\{g_1, \dots, g_k\} = G$ (i.e. $k = n$), then g is exactly uniform after one roll, the sum is 0 for all m and no b works. The original Putnam problem assumes $\{g_1, \dots, g_k\} \subsetneq G$; the hints use this.

Hint 1. Use Fourier analysis on G. For a character χ put $\lambda_\chi = \frac{1}{k}\sum_i \chi(g_i)$; the m-fold convolution gives $\sum_x \Pr(g = x)\overline{\chi(x)} = \lambda_\chi^m$ (up to conjugation), and $\lambda_{\mathbf{1}} = 1$.

Hint 2. Parseval: $\sum_x \left(\Pr(g = x) - \frac{1}{n}\right)^2 = \frac{1}{n}\sum_{\chi \neq \mathbf{1}} |\lambda_\chi|^{2m}$. Take $b = \max_{\chi \neq \mathbf{1}} |\lambda_\chi|$; then the limit is $\frac{1}{n}\#\{\chi : |\lambda_\chi| = b\}$, positive and finite once $0 < b < 1$.

Hint 3. $b < 1$: $|\lambda_\chi| = 1$ forces all $\chi(g_i)$ equal to $\chi(g_1) = \chi(e) = 1$, so χ is trivial on a generating set. $b > 0$: if all λ_χ, $\chi \neq \mathbf{1}$, vanished, the indicator of $\{g_i\}$ would have only a constant Fourier component, i.e. $\{g_i\} = G$.

Problem 3

Hint 1. "At random" means natural density: n uniform on $\{1, \dots, N\}$, $N \to \infty$. Rewrite the parity condition: $\lfloor n\sqrt{2} \rfloor$ is even iff $\{n\sqrt{2}/2\} < \frac{1}{2}$.

Hint 2. Since $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$, Weyl's criterion gives equidistribution of $(n\sqrt{2}/2,\ n\sqrt{3}/2)$ in $[0, 1)^2$. The density is the area of $[0, \frac{1}{2})^2$, i.e. $\frac{1}{4}$.

Problem 4

Hint 1. Shearer: if each coordinate $i \in [n]$ belongs to at least t of the sets $F_1, \dots, F_m$, then $tH(X_1, \dots, X_n) \leqslant \sum_j H(X_{F_j})$. Expand each $H(X_{F_j})$ by the chain rule in increasing order of indices and use that conditioning on fewer variables does not decrease entropy.

Hint 2. For $F = \{i_1 < \cdots < i_s\}$: $H(X_F) = \sum_l H(X_{i_l} \mid X_{i_1}, \dots, X_{i_{l-1}}) \geqslant \sum_l H(X_{i_l} \mid X_1, \dots, X_{i_l - 1})$. Sum over j and compare with the chain rule for $H(X)$.

Hint 3. Loomis–Whitney: for a finite $S \subset \mathbb{Z}^3$ take (X, Y, Z) uniform on S and the three coordinate pairs ($t = 2$): $2\log|S| \leqslant \log|S_{xy}| + \log|S_{yz}| + \log|S_{xz}|$. Apply this to the cubes of a fine grid meeting the body and let the mesh tend to 0.

Problem 5

Hint 1. Take $G(n, p)$ with $p = n^{\theta - 1}$, $0 < \theta < 1/g$. Bound the expected number X of cycles of length $\leqslant g$ and, by a union bound, the probability that the independence number α is large.

Hint 2. $\mathrm{E}X \leqslant \sum_{i=3}^{g}(np)^i = O(n^{\theta g}) = o(n)$, so $\Pr(X \geqslant n/2) \to 0$ by Markov. With $a = \lceil \frac{3}{p}\ln n \rceil$, $\Pr(\alpha \geqslant a) \leqslant \binom{n}{a}(1-p)^{\binom{a}{2}} \leqslant \left(ne^{-p(a-1)/2}\right)^a \to 0$.

Hint 3. For large n pick a graph with $X < n/2$ and $\alpha < a$; delete one vertex from each short cycle. The rest has $\geqslant n/2$ vertices, girth $> g$ and $\chi \geqslant \frac{n/2}{a} \sim \frac{n^\theta}{6\ln n} > k$.

Problem 6

Hint 1. Let $\mathcal{F}$ be an intersecting family of k-subsets of $[n]$, $n \geqslant 2k$. Place $1, \dots, n$ in a uniformly random cyclic order and count the members of $\mathcal{F}$ that are arcs (sets of k consecutive elements).

Hint 2. Lemma: at most k of the n arcs can lie in $\mathcal{F}$. If the arc A is in $\mathcal{F}$, the other arcs meeting A start or end at one of the $k-1$ inner "gaps" of A; for each gap the two arcs (one ending, one starting there) are disjoint when $n \geqslant 2k$, so at most one of them is in $\mathcal{F}$.

Hint 3. Each fixed k-set is an arc with probability $n/\binom{n}{k}$, so $|\mathcal{F}| \cdot n/\binom{n}{k} \leqslant k$, i.e. $|\mathcal{F}| \leqslant \binom{n-1}{k-1}$. For $k > n/2$ any two k-sets meet, so $\binom{n}{k}$ sets form an intersecting family; in the lemma the paired arcs are no longer disjoint.

Problem 7

Hint 1. Measure preservation gives $\mu(T^{-n}A \triangle T^{-(n-1)}A) = \mu(T^{-(n-1)}(T^{-1}A \triangle A)) = 0$, so $T^{-n}A = A$ a.e. for every n.

Hint 2. Take $B = \limsup_n T^{-n}A = \bigcap_N \bigcup_{n \geqslant N} T^{-n}A$. Then $T^{-1}B = \bigcap_N \bigcup_{n \geqslant N} T^{-(n+1)}A = B$ exactly, and $B = A$ a.e. since it is built from countably many sets each a.e. equal to A.

Problem 8

Hint 1. Expand $(\det A)^2 = \sum_{\sigma,\tau} \operatorname{sgn}\sigma \operatorname{sgn}\tau \prod_i a_{i\sigma(i)}a_{i\tau(i)}$. If $\sigma \neq \tau$, some entry occurs exactly once and the expectation vanishes. Hence $\mathrm{E}(\det A)^2 = n!$.

Hint 2. By Markov, $\Pr(|\det A| \geqslant \lambda\sqrt{n!}) \leqslant \lambda^{-2}$: typically $|\det A| = O(\sqrt{n!})$. (The matching lower bound $|\det A| = \sqrt{n!}\, e^{o(n)}$ with high probability is a deep theorem of Tao–Vu; the second moment alone only gives the upper bound.)

Hint 3. Hadamard: $|\det A| \leqslant \prod \|\text{rows}\| = n^{n/2}$, and by Stirling $\sqrt{n!}/n^{n/2} \approx (2\pi n)^{1/4}e^{-n/2}$: a random sign matrix is exponentially far from Hadamard's bound.

Problem 9

Hint 1. Orient each edge of K_n independently and uniformly. For a fixed k-set K, each of the other $n-k$ players beats all of K with probability 2^{-k}, independently.

Hint 2. So $\Pr(\text{some } K \text{ fails}) \leqslant \binom{n}{k}(1-2^{-k})^{n-k}$, and the property exists once this is < 1. Since $\binom{n}{k} \leqslant n^k$ and $(1-2^{-k})^n \leqslant e^{-n2^{-k}}$, it suffices that $n2^{-k} > k\ln n + O(1)$, i.e. $n \approx k^2 2^k \ln 2\,(1+o(1))$.

Problem 10

Hint 1. Start from the weak bound $\mathrm{cr}(G) \geqslant e - 3n$ (for $n \geqslant 3$): delete one edge per crossing until planar and use $e \leqslant 3n-6$ for planar graphs.

Hint 2. Keep each vertex independently with probability p and apply the weak bound to the induced subgraph of a drawing with $\mathrm{cr}(G)$ crossings: $p^4\,\mathrm{cr}(G) \geqslant p^2 e - 3pn$. Choose $p = 4n/e \leqslant 1$: $\mathrm{cr}(G) \geqslant \frac{e}{p^2} - \frac{3n}{p^3} = \frac{e^3}{64n^2}$.

Hint 3. Szemerédi–Trotter: for m points and L lines with I incidences, draw the graph on the points whose edges join consecutive points on each line: $e = I - L$ edges and at most L^2 crossings. Either $e < 4m$, or $L^2 \geqslant \frac{(I-L)^3}{64m^2}$; in both cases $I = O(m^{2/3}L^{2/3} + m + L)$.